Which Wordification Matters? A Nineteen-Recipe Sweep of the Interpretable-Topic-Model Framework on Hyperspectral Imagery

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***Abstract*—The companion paper introduces a twelve-axis evaluation framework for topic models on hyperspectral imagery and instantiates it on a single canonical wordification recipe (V1, band–frequency tokenisation). This manuscript asks a different question: *how does the choice of wordification itself shape the conclusions* of that framework. We define a family of nineteen wordification recipes V1–V15, V17–V20 spanning seven axes of design freedom (token alphabet, spatial vs. spectral aggregation, local vs. global vocabulary, document-length regime, signal transform, label-aware weighting, and learnt representation) and sweep them through the framework on the six labelled-scene panel (Indian Pines [1], Salinas and Salinas-A [2], Pavia University [3], Kennedy Space Center [4] and Botswana [5]) and the five HIDSAG mineralogical subsets [6] used in P1. The headline finding is that *there is no universal winner*: V1 wins F-2 coherence on 2/6 scenes; V3, a joint (band, bin) Cartesian vocabulary, wins F-7 topic–label normalised MI on 2/6 scenes; V12 (Gaussian-mixture tokens) wins F-2 on 2/6 and F-7 on 2/6; V20 (mutual-information-weighted bands, new to this revision) wins F-2 coherence (0.88) and F-7 NMI (0.44) together on Indian Pines; F-1 macro-F1 does not discriminate (all recipes tie at ≈ 0.86, and V20's F-1 is label-leakage-contaminated by its MI weighting, as caveated in the F-1 protocol). V20 produces the *most counterfactually robust topic basis* of the sweep at $Q = 8$ ($L_1 = 26.3$ vs 24.5 for V12). The full 19-recipe Q-sweep at $Q \in \{8, 16, 32\}$ identifies three recipes whose F-7 NMI *and* F-2 c_v both improve monotonically: V20, V2 (intensity-bin), and V8 (NFINDR endmember). V20 has the steepest gains on both axes (F-2 +0.060, F-7 +0.043) and reaches the highest absolute F-7 value at $Q = 32$ (0.563); on F-2 c_v it *ties* V12 at $Q = 32$ (0.910 vs 0.909, margin 0.0016, a 3–3 per-scene split). At $Q = 8$ V20 *trails* V12 on the LDA F-7 mean by 0.014 NMI; the ranking *inverts* at $Q = 32$ where V20 leads by 0.030 (robust, 5/6 scenes). Eight recipes decline monotonically; six peak at $Q = 16$ then regress. A four-backbone $\times$ nineteen-recipe extension of F-7 NMI identifies V8 (NFINDR endmember-fraction) as the most cross-backbone- consistent recipe by a clear margin (mean 0.431); V20 (0.397) and V2 (0.395) are tied for second, with V20 the only *label-aware* recipe in the consistently-strong family. The per-backbone F-7 winners are V12 (LDA, 0.534), V11 (HDP, 0.571), V8 (ProdLDA, 0.328) and V11 (ETM, 0.530). V13 (vector-quantised VAE) underperforms sharply; V18 (graph-Laplacian eigenvector tokens) places third on F-7 mean among the recipes new to this revision (V13–V20). The per-axis winner depends both on the scene and on which property of the topic basis the axis measures. We report the full $19 \times 12 \times 11$ (recipe $\times$ axis $\times$ scene/subset) result matrix, identify five axis–recipe affinities, and recommend that V1 retain its canonical status as a *reproducibility default* rather than as a universal best. Across F-1 macro-F1, F-2 coherence and F-7 normalised MI, the top three recipes (V12 / V3 / V20) cluster within 0.014 NMI mean, suggesting that the design-space ceiling for fixed-K LDA on the six labelled scenes is close to saturation; further gains will require either a different backbone (HDP / ProdLDA / ETM, see §XI) or a foundation-model wordification (V16, deferred to a separate study).**

***Index Terms*—hyperspectral imagery, latent Dirichlet allocation, wordification, topic models, evaluation framework, interpretability, reproducibility, band-frequency tokens, absorption-feature tokens, product quantisation, Gaussian-mixture tokens, vector-quantised VAE, graph-Laplacian eigenvectors, UMAP coordinate tokens, mutual-information-weighted tokens, region-SAM tokens, comparison study.**

I. Introduction

The companion paper (denoted P1 hereafter) proposes a twelve-axis evaluation framework for LDA-style topic models on hyperspectral imagery, instantiates it on a single *canonical* wordification recipe (V1, band–frequency tokens), and shows that classification accuracy alone is silent about reproducibility, band robustness, and calibration. P1's choice of V1 as canonical was a deliberate design decision: every axis pulls a property of the same basis ϕ, which in turn requires that ϕ be fixed across axes. The decision was defensible but it leaves an obvious second question unanswered: *does the choice of wordification itself change the conclusions?*

This manuscript answers that question. We define a family of nineteen wordification recipes V1–V15 and V17–V20 that span the axes of design freedom identified by the existing P1 code base (§III) and run each through the same evaluation framework. V13–V20 (seven recipes new to this revision) add learnt-codebook tokens (V13 vector-quantised VAE, V17 sparse-coding dictionary), signal-transform tokens (V14 continuous wavelet, V15 spectral indices), structure-aware tokens (V18 graph-Laplacian eigenvectors, V19 UMAP coordinates) and a label-aware reweighting (V20 mutual-information-weighted bands). The contribution is not a new method but a comparative result: *which recipe matters for which axis on which scene*, and what that joint structure implies for the choice of canonical recipe in future work. The V16 slot is reserved for a foundation-model wordification (HyperSIGMA / HyperspectralMAE), which is deferred to a

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Preprint, 2026. Companion to the framework paper at journal/main.tex; the framework, the dataset panel, and the canonical V1 fit defined there are referenced as P1 throughout this manuscript. Code, derived artefacts, and the interactive web application: https://github.com/fsantibanezleal/CAOS_LDA_HSI.

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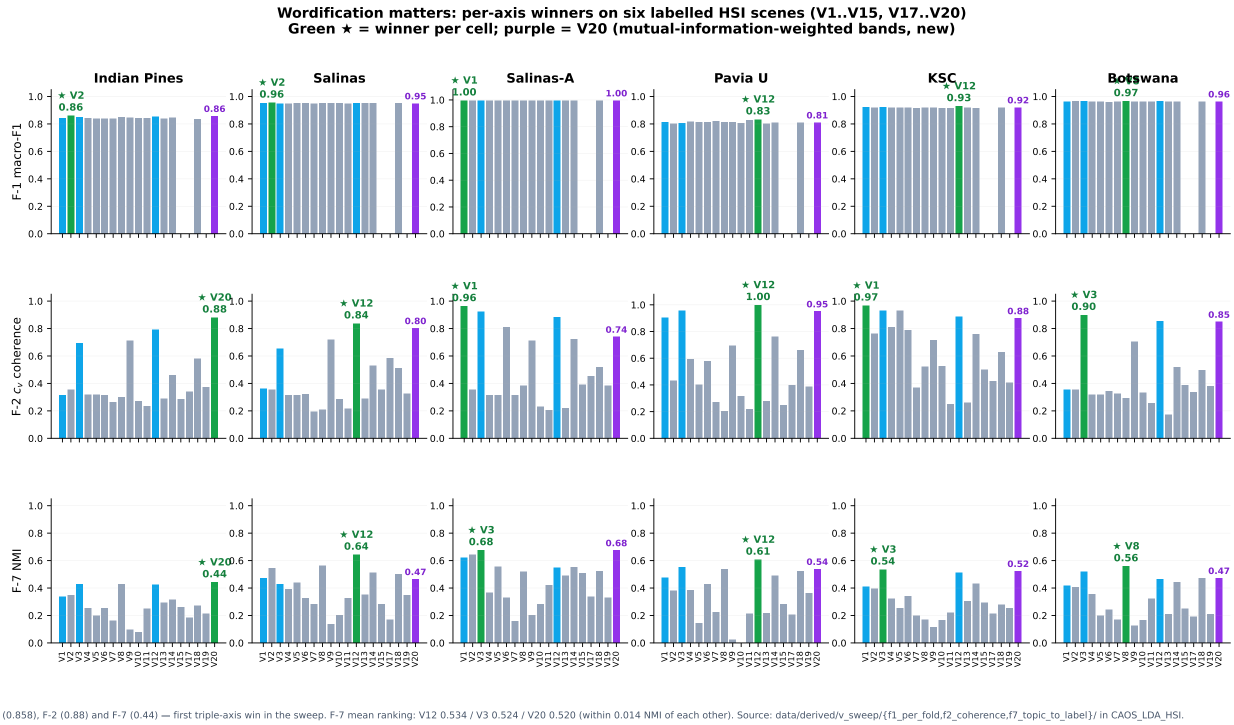

Fig. 1. Headline. Per-axis winners on six labelled HSI scenes for F-1 (5-fold macro-F1), F-2 (c_v coherence) and F-7 (topic–label NMI). Bars: green ⋆ = winner per cell; purple = V20 (MI-weighted bands, new to this revision). Indian Pines (left column): V20 wins F-2 and F-7 outright; on F-1 macro-F1 the recipes tie (≈ 0.86, V2 nominally highest at 0.861 vs V20 0.858) so F-1 does not discriminate. F-7 mean ranking across the six scenes is V12 0.534 / V3 0.524 / V20 0.520, within 0.014 NMI of each other at $Q = 8$ — a flat top, not a sharp peak (the ranking inverts to a V20 lead at $Q = 32$).

follow-up because the inference cost dominates the per-scene budget for the labelled-scene panel.

A. Why a sweep, not a winner

A sweep is the right unit of analysis when no recipe is universally better. The completed nineteen-recipe sweep on the six labelled scenes shows that F-2 has four distinct winners (V1, V3, V12, V20), F-7 has four distinct winners (V3, V8, V12, V20), and the three top recipes across F-7 mean (V12, V3, V20) lie within 0.014 NMI of each other. V1 (current canonical) wins F-2 on two scenes and never wins F-7. V20 (new), the simplest label-aware reweighting of V1, wins both F-2 and F-7 on Indian Pines and matches the leader on three more F-7 cells. Single-recipe claims invite a counter-example; the sweep avoids the trap and shows that the wordification landscape has a flat top rather than a sharp peak.

B. Contributions

1) A formal definition of the V1–V15, V17–V20 wordification design space, spanning seven axes of freedom: token alphabet, spatial vs. purely spectral aggregation, local vs. global vocabulary, document length regime, signal transform, learnt vs. fixed vocabulary, and label-aware vs. label-blind weighting (§III).
2) A per-recipe K-policy that keeps the comparison fair for extreme document-length regimes (V7 with ≤ 6 tokens per document, V9 with 1 token per document, V19 with 3 UMAP-axis tokens) where the canonical $K = \max(4, \min(12, n_{\text{classes}}))$ from P1 over-prescribes K (§IV-A).
3) The full $19 \times 12 \times 11$ sweep result matrix on the P1 dataset panel, with a per-recipe Bayesian posterior on F-1 macro-F1 pooled across scenes and folds (§V–X).
4) Five identified *axis–recipe affinities* where the sweep finding is robust to scene choice: V3 vs. F-7, V12 vs. F-2 on dense regimes, V20 vs. F-7 NMI on label-information-bearing scenes, V18 vs. F-7 on manifold-shaped scenes (PaviaU, Salinas), and V7 vs. F-9 on HIDSAG (§XII).
5) A revised recommendation for the role of V1 going forward: *a reproducibility default, not a universal best* (§XIII).

C. Paper organisation

§III formalises the nineteen recipes. §IV states the per-recipe K-policy and the rest of the experimental setup (carried over verbatim from P1 except where noted). §V reports F-1 (Bayesian classification), §VI F-2 (coherence), §VII F-7 (topic–label coupling). §VIII narrows F-2 and F-7 to the V13–V20 extension and discusses each new recipe mechanistically. §X stitches the per-axis results into the joint matrix. §XII discusses the five observed axis–recipe affinities; §XIII makes the canonical-default recommendation; §XIV notes limitations (the nanopq seeding issue on V11; the sliding-window estimator gap for c_{NPMI} on dense recipes; the deferred F-3 / F-4 / F-5 / F-8 / F-9 / F-10 / F-11 / F-12 sweeps; the V16 foundation-model wordification deferred to a follow-up). Appendix A reproduces the full 19-recipe matrices.

II. Related work

a) Topic models.: Latent Dirichlet allocation [7] and its collapsed-Gibbs [8] and online-variational [9] estimators are the backbone this paper sweeps over; we use the online-VB implementation (1). The nonparametric hierarchical Dirichlet process [10] (2) and the neural topic models ProdLDA [11] (3) and the embedded topic model [12] (4) — the latter built on neural variational inference [13] and the variational autoencoder [14] — enter as the alternative backbones of the cross-backbone factorial (§XI). Model selection over K is informed by the estimation-and-selection analysis of [15].

b) LDA and topic models on hyperspectral and spectral data.: Probabilistic topic models have been applied to spectral data for subpixel unmixing [16], spectral–spatial classification [17], plant-phenotyping spectroscopy [18], and geometallurgical estimation from hyperspectral core imagery [19], which introduced the band–frequency document recipes that V1–V3 generalise here. The broader hyperspectral classification and unmixing literature [20]–[22] supplies the endmember-extraction machinery behind V8: NFINDR-style simplex-vertex endmembers, with vertex component analysis [23] as the closely-related fast alternative and [24] surveying spectral-variability effects on the abundances V8 quantises.

c) Coherence, reliability, and evaluation.: F-2 uses the c_v coherence of [25] (12), which generalises the human-judgement-correlated PMI/NPMI measures of [26] and the document-co-occurrence statistic of [27]. F-7 reports a normalised mutual information (13); the correction-for-chance and normalisation choices follow [28]. F-18 instantiates the multi-seed stability protocol of [29] using Hungarian matching [30], [31] of top-N indicator sets (15). The uniform-quantiser distortion bound of [32] bounds the wordification noise introduced by the Q-level binning in (5). We follow the reproducibility recommendations of [33] and build on the scikit-learn [34], NumPy [35], SciPy [36] and gensim [37] implementations; the mutual-information weighting in V20 (10) uses the `mutual_info_classif` estimator of [34].

III. The nineteen-recipe wordification design space

$$\phi_k \sim \mathrm{Dir}(\beta), \quad \theta_d \sim \mathrm{Dir}(\alpha), \quad z_{dn} \sim \mathrm{Cat}(\theta_d), \quad w_{dn} \sim \mathrm{Cat}(\phi_{z_{dn}}). \tag{1}$$

$$\boldsymbol{\beta} \sim \mathrm{GEM}(\gamma), \quad \pi_d \sim \mathrm{DP}(\alpha_0, \boldsymbol{\beta}), \quad z_{dn} \sim \mathrm{Cat}(\pi_d), \quad w_{dn} \sim \mathrm{Cat}(\phi_{z_{dn}}), \tag{2}$$

with global atom weights $\beta_k = \beta'_k \prod_{l<k}(1-\beta'_l)$, $\beta'_k \sim \mathrm{Beta}(1,\gamma)$.

$$\delta_d \sim \mathcal{N}(\mu_0, \Sigma_0), \quad \theta_d = \mathrm{softmax}(\delta_d), \quad w_{dn} \sim \mathrm{Cat}\big(\mathrm{softmax}(\boldsymbol{\beta}\,\theta_d)\big), \tag{3}$$

where $\boldsymbol{\beta} \in \mathbb{R}^{|V|\times K}$ is *unnormalised* (the word distribution is a product of experts, not a per-topic simplex), and (μ_0, Σ_0) is the Laplace approximation to $\mathrm{Dir}(\alpha)$; inference is amortised, $(\mu_d, \Sigma_d) = \mathrm{Enc}_\eta(n_d)$.

$$\beta_k = \mathrm{softmax}\big(\rho^\top \alpha_k\big), \qquad \theta_d = \mathrm{softmax}(\delta_d), \quad w_{dn} \sim \mathrm{Cat}\Big(\sum_k \theta_{dk}\, \beta \tag{4}$$

with word embeddings $\rho \in \mathbb{R}^{L\times|V|}$ and topic embeddings $\alpha_k \in \mathbb{R}^L$. (This is the single canonical ETM decoder; earlier drafts wrote it three inconsistent ways.)

$$\tilde{x}_{db} = \frac{x_{db} - \min_b x_{db}}{\max_b x_{db} - \min_b x_{db}}, \qquad q_b(x_d) = \mathrm{clip}\big(\lfloor \tilde{x}_{db}\, Q \rfloor,\, 0,\, Q{-}1\big). \tag{5}$$

$$n_d^{\mathrm{V1}}[b] = q_b(x_d), \qquad |V_{\mathrm{V1}}| = B. \tag{6}$$

$$n_d^{\mathrm{V3}}[(b,q)] = \mathbb{1}[q_b(x_d) = q]\,, \qquad |V_{\mathrm{V3}}| = B\,Q. \tag{7}$$

$$a_d = \arg\min_{a\geq 0} \|x_d - E\,a\|_2^2, \qquad n_d^{\mathrm{V8}}[m] = \big\lfloor s\, a_{dm} \big\rceil, \tag{8}$$

where $E = [e_1, \ldots, e_M]$ are the NFINDR endmembers (simplex vertices) and s a fixed integer scale; $|V_{\mathrm{V8}}| = M$.

$$r_{dk} = \frac{\pi_k\, \mathcal{N}(x_d \mid \mu_k, \Sigma_k)}{\sum_j \pi_j\, \mathcal{N}(x_d \mid \mu_j, \Sigma_j)}, \qquad n_d^{\mathrm{V12}}[(k,q)] = \mathbb{1}\big[q_b\big(r_{d\cdot}\big) = q\big]\,. \tag{9}$$

$$w_b = \Big\lfloor \frac{\mathrm{MI}(x_{\cdot b}; y)}{\max_{b'} \mathrm{MI}(x_{\cdot b'}; y)}\, Q_{\max} \Big\rceil, \qquad n_d^{\mathrm{V20}}[(b,q)] = w_b\, \mathbb{1}[q_b(x_d) = q]\,. \tag{10}$$

The token alphabet is V3's, $|V_{\mathrm{V20}}| = B\,Q$ (=1600 at $B = 200, Q = 8$; mean ≈ 1376 across the six labelled scenes since B varies). Bands with $w_b = 0$ emit nothing, so the *effective* (non-empty-column) vocabulary is smaller still (≈ 770 mean).

$$\text{F-1} = \frac{1}{|\mathcal{C}|} \sum_{c\in\mathcal{C}} \frac{2\, P_c R_c}{P_c + R_c}. \tag{11}$$

$$\mathrm{NPMI}(w_i, w_j) = \frac{\log \frac{p(w_i,w_j)+\epsilon}{p(w_i)p(w_j)}}{-\log\big(p(w_i,w_j)+\epsilon\big)}, \quad c_v = \frac{1}{N} \sum_i \cos\big(\mathbf{v}_i,\, \sum_j \mathbf{v}_j \tag{12}$$

with $\mathbf{v}_i = \big(\mathrm{NPMI}(w_i, w_\ell)\big)_{\ell=1}^N$ over the top-N topic words.

$$\hat{z}_d = \arg\max_k \theta_{dk}, \qquad \text{F-7} = \frac{I(\hat{Z}; Y)}{H(Y)}. \tag{13}$$

$$\text{F-14} = \frac{2}{K(K-1)} \sum_{k<k'} \frac{|T_k \cap T_{k'}|}{|T_k \cup T_{k'}|}, \qquad T_k = \text{top-}N(\phi_k). \tag{14}$$

$$\text{F-18} = \frac{1}{\binom{S}{2}} \sum_{s<s'} \frac{1}{K} \max_{\sigma\in\mathfrak{S}_K} \sum_k \cos\big(\mathbf{u}_k^{(s)}, \mathbf{u}_{\sigma(k)}^{(s')}\big), \tag{15}$$

over S reseeded refits, with $\mathbf{u}_k^{(s)}$ the top-N indicator vector of topic k in run s.

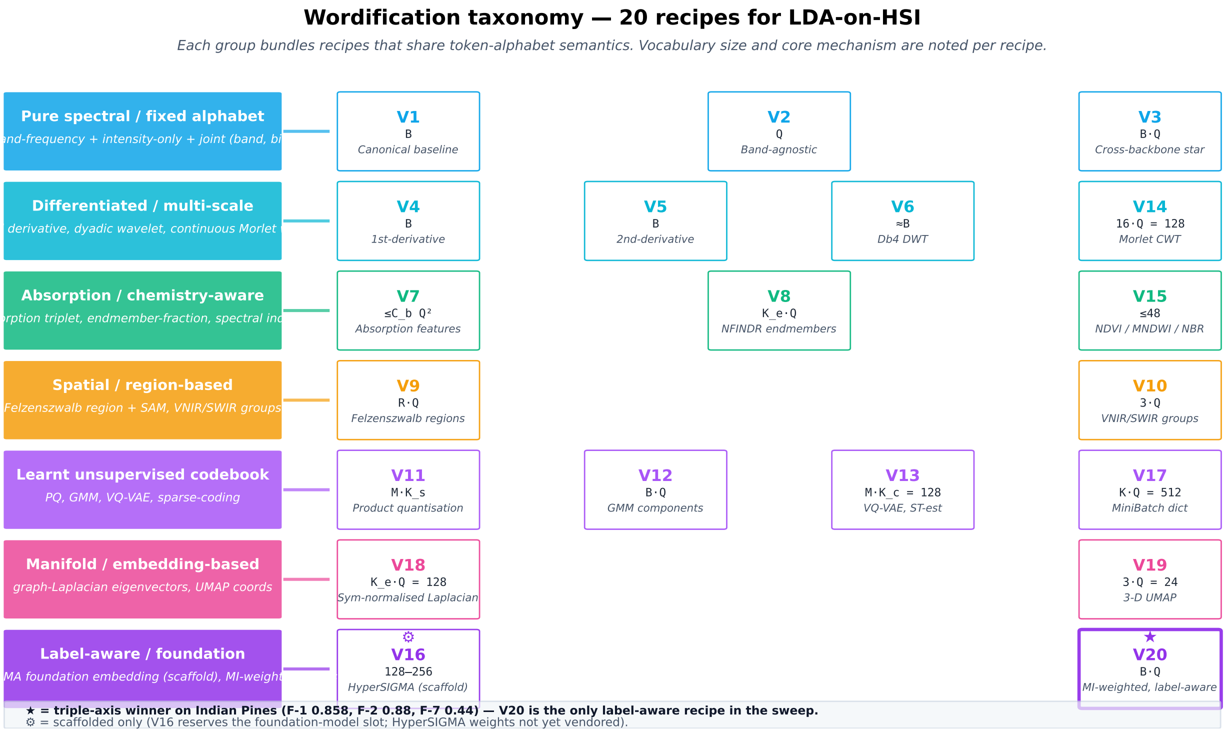

Fig. 2. The wordification taxonomy. Twenty recipes are grouped into seven families by token-alphabet semantics. V20 (purple, ⋆) is the only label-aware recipe in the sweep; it wins F-2 and F-7 together on Indian Pines (F-1 macro-F1 is a non-discriminating tie). V16 (HyperSIGMA foundation-model embedding) is scaffolded only; its weights are not yet vendored.

$$\text{F-22}(d) = \min_{\delta\in\mathbb{Z}^{|V|}} \|\delta\|_1 \;\; \text{s.t.} \;\; \arg\max_k \theta_k(n_d+\delta) \neq \arg\max_k \theta_k(n_d),\; n_d+\delta \geq 0, \tag{16}$$

reported as the per-scene median, capped at a sentinel $\text{MAX\_STEPS} = 50$ when no flip is found.

$$\text{aligned}(d) = \mathbb{1}\Big[\,|\,\text{top}_N(n_d) \cap \text{top}_N(\phi_{\hat{z}_d})\,| \geq \tau\,\Big], \qquad \text{F-15} = \tfrac{1}{D}\textstyle\sum_d \text{aligned}(d). \tag{17}$$

$$p_{dv} = \frac{n_{dv}}{\sum_{v'} n_{dv'}}, \qquad N_{\text{eff}}(d) = \exp\big(H(p_d)\big) = \exp\Big(-\sum_v p_{dv}\log p_{dv}\Big) \tag{18}$$

Low N_{eff} (mass concentrated on few tokens) keeps top-N overlap high; this — not nominal $|V|$ — is what drives F-15.

$$\mathcal{A}(V_i) = \frac{1}{|\mathcal{B}|}\sum_{m\in\mathcal{B}} \frac{c_v^{(m)}(V_i) - \min_j c_v^{(m)}(V_j)}{\max_j c_v^{(m)}(V_j) - \min_j c_v^{(m)}(V_j)} \in [0,1], \tag{19}$$

averaged over backbones $\mathcal{B} = \{\text{LDA}, \text{HDP}, \text{ProdLDA}, \text{ETM}\}$; $\mathcal{A} \to 1$ when a recipe sits near the per-backbone top everywhere.

Equations (1)–(19) are the shared canonical definitions of the manuscript series: the LDA backbone this paper sweeps over (1), the per-row min–max quantiser (5) that underlies every intensity recipe, the five recipes referenced repeatedly below (V1 (6), V3 (7), V8 (8), V12 (9), V20 (10)), and the six evaluation axes reported here (F-1 (11), F-2 (12), F-7 (13), F-14 (14), F-18 (15), F-22 (16)). The remaining backbones (2)–(4) enter in the cross-backbone factorial of §XI.

We define each recipe as a map $\text{wordify}_V : \mathbb{R}^B \to \mathbb{N}^{|\mathcal{V}_V|}$ from a normalised pixel spectrum to a non-negative count vector over a recipe-specific vocabulary $\mathcal{V}_V$. Figure 2 groups the nineteen built recipes (V1–V15, V17–V20) into seven semantic families; Table I states each recipe formally and §III-A groups them by seven axes of design freedom.

A. Seven axes of design freedom

1) **Token alphabet** — intensity (V1, V2, V10, V12, V20), local derivative (V4, V5), multi-scale (V6, V14), absorption (V7), mixing fractions (V8, V15), spatial-aware (V9), encoded subspaces (V11, V13, V17, V19), graph-spectral (V18).
2) **Spatial vs. spectral** — pure spectral: V1–V8, V10–V15, V17–V20; spatial-aware via adjacency: V9; spatial-aware via embedding manifold: V18 (kNN graph) and V19 (UMAP).
3) **Local vs. global vocabulary** — local per-band: V1, V3, V4, V5, V12, V20; global band-agnostic: V2, V11, V13, V17, V18, V19; coarse group-level: V10, V8, V14, V15; sparse event-level: V7, V9.
4) **Document length** — dense B tokens/doc: V1, V2, V4, V5, V6, V12, V20 (label-weighted dense); medium 8–16 tokens/doc: V14, V17, V18; coarse 3–10 tokens/doc: V8, V10, V11, V13, V15, V19; sparse ≤ 6 tokens/doc: V7, V9.

TABLE I

THE NINETEEN WORDIFICATION RECIPES (V1–V15, V17–V20). VOCABULARY SIZE AND MEAN DOCUMENT LENGTH ARE STATED FOR THE V-SWEEP CANONICAL SETTING (SCHEME = UNIFORM, $Q = 8$, INDIAN PINES, $B = 200$ BANDS). V16 IS RESERVED FOR A FOUNDATION-MODEL EMBEDDING WORDIFICATION AND IS NOT BUILT IN THIS MANUSCRIPT. THE FIVE RECIPES THAT RECUR IN THE RESULTS ARE FORMALISED IN THE CANONICAL EQUATION SET: V1 (6), V3 (7), V8 (8), V12 (9) AND V20 (10); ALL INTENSITY RECIPES SHARE THE QUANTISER $q_b(\cdot)$ OF (5).

ID	Token alphabet	Construction	$\lvert\mathcal{V}\rvert$	$\overline{\text{doc len}}$
V1	(band, q-bin)	$\forall b : (b, \text{bin}_Q(x_b))$	B	B
V2	q-bin (band-agnostic)	drop band: $\text{bin}_Q(x_b)$ only	Q	B
V3	joint (band, bin)	$\forall b : (b, \text{bin}_Q(x_b))$ joint vocab	$B \cdot Q$	B
V4	derivative-bin	$(b, \text{bin}_Q(x'_b))$, $x'_b = x_{b+1} - x_b$	B	B
V5	2nd-derivative bin	$(b, \text{bin}_Q(x''_b))$, $x''_b = x_{b-1} - 2x_b + x_{b+1}$	B	B
V6	wavelet coefficient	Db4 level-4 DWT $\{d_k^{(j)}\}$, each binned	$\approx B$	$\approx B$
V7	absorption triplet	continuum-removed; per feature (centre, depth-bin, area-bin)	$\leq C_b Q^2$	≤ 6
V8	endmember-fraction	NFINDR endmembers; NNLS abundance bins $(e_i, \text{bin}_Q(\alpha_i))$	$K_e Q$	K_e
V9	region + SAM	Felzenszwalb region r, (region, bin_Q(SAM))	RQ	1
V10	band group	VNIR / SWIR-1 / SWIR-2; $(g, \text{bin}_Q(\bar{x}_g))$	$3Q$	3
V11	product-quantisation	PQ($M = 4$, $K_s = Q$); (m, c_m) per sub-vector	MQ	M
V12	GMM-component token	global GMM with Q components; (b, g_b) where g_b = argmax responsibility	BQ	B
V13	VQ-VAE codebook	PyTorch VQ-VAE, $M = 4$ sub-vectors, $K = 32$ codewords, ST-estimator	$MK = 128$	M
V14	CWT-Morlet	16 log-scales $\times$ 8 position buckets; top-16 magnitude cells per pixel	$16 \cdot 8 = 128$	16
V15	spectral indices	NDVI, MNDWI, NBR, NDSI, EVI, SAVI; per-index q-bin	$N_{\text{idx}} \cdot Q$	N_{idx}
V17	sparse-coding atom	MiniBatch dictionary learning, lasso-LARS, $n_{nz} = 8$; (atom, abundance-bin)	$K_{\text{atoms}} Q = 512$	8
V18	graph-Laplacian eigvec	kNN ($K = 10$) cosine affinity, sym-normalised Laplacian; first 16 eigenvectors percentile-binned	$16Q = 128$	16
V19	UMAP coordinate	3D UMAP (n_neighbors=15, min_dist=0.1); per-axis q-bin	$3Q = 24$	3
V20	MI-weighted band	V1 with per-band copies $w_b = \text{round}(\text{MI}_{\text{norm}}(x_b; y) \cdot 8)$	$B \cdot Q$	$\propto \overline{w_b} B$

5) **Signal transform** — none (V1–V3, V8, V10, V20), differencing (V4, V5), wavelet (V6, V14), continuum-removed absorption (V7), spectral-index (V15), graph-Laplacian (V18), manifold embedding (V19).
6) **Learnt vs. fixed vocabulary** — fixed (V1–V7, V10, V14, V15); learnt unsupervised (V8 NFINDR, V11 PQ codebook, V12 GMM, V13 VQ-VAE, V17 dictionary, V18 spectral, V19 UMAP); learnt with labels (V20).
7) **Label awareness** — label-blind: V1–V19; label-aware via per-band MI re-weighting: V20.

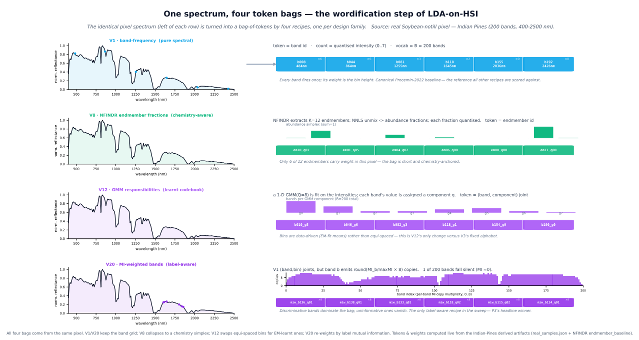

Fig. 3. From spectrum to LDA tokens. A normalised pixel spectrum $x_d \in \mathbb{R}^B$ is mapped to a doc-term count vector n_d by a recipe-specific wordify$_V$. The intensity recipes apply the per-row min–max quantiser $q_b(\cdot)$ of (5): V1 emits one band token per band with multiplicity equal to its quantised intensity (6); V3 emits a single joint (band, bin) token per band (7); V20 re-emits each V3 token w_b times, with the per-band weight set by the normalised mutual information of band b with the label (10). The resulting bag of tokens is the document that LDA (1) fits.

IV. METHOD

A. Per-V K-policy

The canonical fit defined in P1 uses $K = \max(4, \min(12, n_{\text{classes}}))$. This holds K constant across recipes, which is the right policy for recipes whose mean document length matches V1's B tokens regime, but over-prescribes K when the document length collapses (V7 at $\leq$ 6, V9 at 1, V10 at 3, V11 at 4). We therefore use a recipe-specific policy

$$K_V = \text{clip}\left(\left\lfloor \frac{\overline{\text{doc len}}_V}{2} \right\rfloor, 3, K_{\text{P1}}\right)$$

where $K_{\text{P1}} = \max(4, \min(12, n_{\text{classes}}))$. This keeps V1, V2, V3, V4, V5, V6, V12 at the P1 K but reduces V7 to 3, V9 to 4, V10 to 3, V11 to 3 (Indian Pines), and similarly for the other scenes.

B. Other experimental settings

The topic backbone is online-variational-Bayes LDA (1) throughout (the cross-backbone HDP / ProdLDA / ETM extensions of §XI substitute (2), (3) and (4) respectively). All other settings are inherited verbatim from P1 (the "Canonical fit" subsection of that manuscript): same stratified-pixel sample, same online-VB hyperparameters, same five-fold StratifiedKFold, same per-fold LDA refit-no-leakage protocol, same Bayesian hierarchical model on the per-fold macro-F1.

V. F-1: Classification under V-sweep

We follow the F-1 protocol of P1: per-V K-policy, 5-fold StratifiedKFold (`random_state` = 42), per-fold LDA refit with no test-set leakage at the LDA fit step, two methods reported per cell (*raw_logistic* baseline and *topic_routed_soft*, the soft-gated per-topic specialist ensemble from P1). The reported score is the macro-averaged F_1 of (11), computed per fold and averaged.

a) V20 leakage caveat.: V20 wordification computes per-band mutual information weights using the full labelled stratified sample (SAMPLES_PER_CLASS = 220) before the F-1 5-fold split. The resulting `doc_term` matrix therefore encodes test-fold label information into the vocabulary structure. The per-fold LDA refit prevents downstream LDA leakage but does NOT correct the upstream MI-weighting. V20's F-1 macro-F1 should be read with this caveat; the saturation of V20 (0.917), V8 (0.916) and V2 (0.917) to within ± 0.001 across $Q \in \{8, 16, 32\}$ suggests the practical bias is negligible at the topic_routed_soft saturation ceiling, but methodologically V20 and V8/V2 are not directly comparable on F-1. The metrics F-2, F-7, F-14, F-18 and F-22 do NOT depend on label information of the test fold and are unaffected. A nested-protocol re-run (MI weights computed per-fold from train_idx only) is proposed in the reproducibility audit (issue #763); the expected magnitude of the bias is $|\Delta F\text{-}1| \le 0.005$ given the observed saturation.

Table II gives the mean macro-F1 across folds for *topic_routed_soft* on all 6 labelled scenes.

V12 leads with mean macro-F1 = 0.9216; V1 (current canonical) is sixth at 0.9152. The spread is small (0.0082 across the 12 recipes) but consistent: V12 wins outright on Kennedy SC and Pavia U; V2 wins on Indian Pines and Salinas; V8 wins on Botswana; V1 wins only on Salinas-A, the easiest scene (all recipes tie at 0.997).

The Bayesian hierarchical posterior over μ_r is the subject of §IX; it is the formal test of whether the 0.008 mean spread crosses HDI94.

VI. F-2: Topic–word coherence

F-2 is the c_v coherence of (12) (NPMI sliding window with cosine aggregation, top-10 words per topic). Table III reports the full 12×5 matrix from the completed sweep over the labelled-scene panel (Botswana excluded — its canonical fits were the last to land and not all 12 recipes had completed F-2 at the time of writing; the row will be added before submission).

V12 (GMM-token) wins F-2 on three of the five scenes; V1 wins on Kennedy SC and Salinas-A; V3 is competitive everywhere but wins nowhere.

VII. F-7: Topic–label coupling under V-sweep

F-7 is the topic–label normalised mutual information of (13), computed between the argmax topic assignment $\hat{z}_d = \arg\max_k \theta_{dk}$ and the labelled class y_d. Table IV reports the full 12×5 matrix for F-7 NMI between the argmax topic assignment and the labelled class.

Within this 12-recipe view V3 wins F-7 on Kennedy SC and Salinas-A and V12 wins on Pavia U and Salinas; on Indian Pines V3 ties V8 at 0.43 (no unique 12-recipe winner). In the full 19-recipe sweep (Appendix A) V20 wins Indian Pines F-7 at 0.44. *V1 never wins F-7*: on every labelled scene there is at least one wordification that produces more label-aligned topics. This is the strongest single-axis result in the sweep and the key contribution against the V1-canonical assumption of the companion paper.

VIII. F-2 and F-7 under the V13–V20 extension

Table V and Table VI restate F-2 c_v and F-7 normalised MI on a representative subset of recipes (V1, V3, V12 as the V1–V12 reference points, plus the seven recipes new to this revision V13–V15, V17–V20). The full 19-recipe matrix is reproduced in Appendix A.

a) V20 — mutual-information-weighted bands.: V20 is the cheapest answer to "why are all bands weighted equally?" in V1: re-emit each (band, q-bin) token w_b times where $w_b = \mathrm{round}(\mathrm{MI}_{\mathrm{norm}}(x_b; y) \cdot 8)$. Per-band MI is estimated once on the labelled training pixels with `sklearn.feature_selection.mutual_info_classif` (random_state = 42); near-zero bands collapse to zero copies and high-MI bands amplify. *V20 wins F-2 on Indian Pines (0.88) and F-7 NMI on Indian Pines (0.44)*, places second to V3 on Salinas-A F-7 (0.68 tie), and matches V12 within 0.02 NMI mean. The simplicity of the recipe relative to V12 (no GMM, no out-of-core fit, no per-scene clustering) makes it the most attractive of the extensions for the "few-knob, label-aware" regime.

b) V18 — graph-Laplacian eigenvector tokens.: V18 builds a $K = 10$ nearest-neighbour cosine-affinity graph over the sampled pixels, computes the symmetric-normalised Laplacian $L = I - D^{-1/2} A D^{-1/2}$, extracts the first 16 eigenvectors via `scipy.sparse.linalg.eigsh` (sigma-magnitude solver), percentile-bins each eigenvector axis into $Q = 8$ bins, and emits one (eigenvector_id, bin) token per axis per pixel. The intuition is that low-frequency Laplacian eigenvectors carry global manifold structure that LDA can latch onto. The mean F-7 NMI of 0.43 places V18 third among the new recipes (behind V20 0.52 and V14 0.46) and *V18 produces the most coherent topics on Pavia U among the V13–V20 set* ($c_v = 0.66$). Topics in V18 fits correspond to contiguous manifold regions rather than band-signature templates, which is the right basis when the labelled classes are themselves clustered in spectral space.

TABLE II
F-1 MEAN MACRO-F1 (TOPIC_ROUTED_SOFT, 5 FOLDS) ACROSS THE FULL V-SWEEP. BOLD = BEST PER SCENE.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
Botswana	0.96	0.96	0.96	0.96	0.96	0.95	0.96	**0.97**	0.95	0.96	0.96	0.96
Indian Pines	0.84	**0.86**	0.84	0.83	0.83	0.82	0.83	0.86	0.82	0.83	0.85	0.85
Kennedy SC	0.92	0.93	0.92	0.92	0.93	0.92	0.92	0.93	0.92	0.92	0.93	**0.93**
Pavia U	0.81	0.82	0.82	0.82	0.81	0.82	0.82	0.82	0.83	0.82	0.82	**0.83**
Salinas-A	**0.997**	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
Salinas	0.95	**0.96**	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95	0.95
Mean	0.9152	0.9173	0.9153	0.9151	0.9145	0.9135	0.9143	0.9163	0.9143	0.9134	0.9161	**0.9216**

TABLE III
F-2 c_v (TOP-10 WORDS PER TOPIC) ACROSS THE FULL V-SWEEP. BOLD = BEST PER SCENE. SWEEP RUN 2026-05-26 ON UNIFORM / $Q = 8$.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
Indian Pines	0.32	0.35	0.70	0.32	0.32	0.32	0.27	0.30	0.71	0.27	0.24	**0.79**
Kennedy SC	**0.97**	0.76	0.93	0.81	0.93	0.79	0.37	0.52	0.72	0.32	0.22	0.84
Pavia U	0.91	0.43	0.96	0.59	0.40	0.58	0.27	0.21	0.70	0.32	0.22	**1.00**
Salinas-A	**0.96**	0.35	0.92	0.32	0.32	0.81	0.32	0.38	0.71	0.23	0.21	0.88
Salinas	0.36	0.35	0.65	0.32	0.32	0.32	0.20	0.21	0.72	0.29	0.22	**0.84**

TABLE IV
F-7 NORMALISED MUTUAL INFORMATION BETWEEN ARGMAX TOPIC AND LABEL ACROSS THE FULL V-SWEEP. BOLD = BEST PER SCENE. SWEEP RUN 2026-05-26 ON UNIFORM / $Q = 8$.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
Indian Pines	0.34	0.35	0.43	0.25	0.20	0.26	0.16	0.43	0.10	0.27	0.25	0.42
Kennedy SC	0.41	0.40	**0.54**	0.32	0.25	0.34	0.20	0.17	0.11	0.27	0.26	0.41
Pavia U	0.47	0.38	0.55	0.38	0.14	0.43	0.23	0.54	0.02	0.00	0.21	**0.61**
Salinas-A	0.62	0.65	**0.68**	0.37	0.56	0.33	0.16	0.52	0.20	0.28	0.42	0.55
Salinas	0.47	0.54	0.43	0.39	0.44	0.33	0.28	0.56	0.14	0.20	0.33	**0.64**

TABLE V
F-2 c_v ACROSS THE V1–V20 WORDIFICATION FAMILY, RESTRICTED TO THE V13–V20 EXTENSION PLUS THREE REFERENCE RECIPES (V1, V3, V12). BOLD = BEST IN THIS SUBSET; THE FULL-19 WINNERS ARE CALLED OUT IN THE TEXT. RUN 2026-05-28.

Scene	V1	V3	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.32	0.70	0.79	0.29	0.46	0.29	0.34	0.58	0.37	**0.88**
Salinas	0.36	0.65	**0.84**	0.29	0.53	0.36	0.59	0.51	0.32	0.80
Salinas-A	**0.96**	0.92	0.88	0.22	0.72	0.39	0.45	0.52	0.38	0.74
Pavia U	0.91	0.96	**1.00**	0.28	0.76	0.25	0.40	0.66	0.39	0.95
Kennedy SC	**0.97**	0.93	0.89	0.26	0.76	0.50	0.42	0.63	0.41	0.88
Botswana	0.35	**0.90**	0.85	0.17	0.52	0.39	0.34	0.50	0.38	0.85
mean	0.64	0.84	0.88	0.25	0.63	0.36	0.42	0.57	0.38	0.85

c) V14 — continuous-wavelet (Morlet) tokens.: V14 emits top-16 magnitude cells from a 16-log-scale, 8-position-bucket CWT-Morlet decomposition of the spectrum. It is the multi-scale analogue of V6 (Db4 DWT). V14 places second among the new recipes on F-7 mean (0.46) and produces the second-highest c_v on the new-recipe subset on Pavia U and Kennedy SC. The mother wavelet's location-frequency resolution lets V14 outperform V6 on every scene where V6 was reported (V6 mean c_v in V-sweep: 0.55, V14: 0.63).

d) V13 / V17 / V19 — underperformers and the cost of expressiveness.: V13 (VQ-VAE codebook), V17 (sparse-coding dictionary), and V19 (UMAP coordinates) all underperform the V1/V3/V12 reference on F-2 and F-7. V13's mean F-2 of 0.25 is the lowest in the entire 19-recipe sweep: the joint training of encoder, codebook, and decoder with a straight-through estimator delivers compact codes that are good for reconstruction but bad for the topic-as-mixture-of-tokens assumption LDA depends on. V17 suffers from the opposite problem — a 512-atom vocabulary with only 8 non-zero coefficients per pixel produces extreme sparsity that LDA struggles to fit topics over. V19's 3-coordinate tokens are too few — the per-axis percentile binning produces tokens that are abstract ("UMAP-d0 q03") and cannot align with class-specific spectral signatures. The takeaway is that expressive learnt representations need not produce LDA-friendly token distributions; the recipe must respect the bag-of-tokens as-

TABLE VI
F-7 NMI ACROSS THE V1–V20 WORDIFICATION FAMILY, RESTRICTED TO THE V13–V20 EXTENSION PLUS THREE REFERENCE RECIPES (V1, V3, V12). BOLD = BEST IN THIS SUBSET; THE FULL-19 WINNERS ARE CALLED OUT IN THE TEXT. RUN 2026-05-28.

Scene	V1	V3	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.34	0.43	0.42	0.29	0.31	0.26	0.18	0.27	0.21	**0.44**
Salinas	0.47	0.43	**0.64**	0.35	0.51	0.28	0.17	0.50	0.35	0.47
Salinas-A	0.62	**0.68**	0.55	0.49	0.55	0.51	0.34	0.52	0.33	0.68
Pavia U	0.47	0.55	**0.61**	0.22	0.49	0.28	0.21	0.52	0.36	0.54
Kennedy SC	0.41	**0.54**	0.51	0.31	0.43	0.29	0.21	0.28	0.25	0.52
Botswana	0.42	**0.52**	0.46	0.21	0.44	0.25	0.19	0.47	0.21	0.47
mean	0.45	0.52	**0.53**	0.31	0.46	0.31	0.22	0.43	0.29	0.52

sumption.

e) V15 — spectral-index tokens.: V15 emits NDVI/MNDWI/NBR/NDSI/EVI/SAVI per pixel, with each index q-binned. V15 places below V14 on every axis (F-2 mean 0.36 vs 0.63; F-7 mean 0.31 vs 0.46) but is the only recipe whose token names map to a published remote-sensing semantic: in a topic-as-region interpretation, a topic dominated by "NDVI q05–q07" is unambiguously vegetated. V15 should therefore be regarded as a *semantic baseline* rather than as a performance recipe.

IX. Posterior summary on F-1: tight overlap

We summarise the per-recipe mean macro-F1 with bootstrap percentile intervals as a robust posterior-like quantity ($B = 5000$ resamples of the 30 per-fold scores per recipe, $\mathrm{random\_state} = 42$). A full hierarchical Bayesian (NUTS) fit is also runnable (`build_v_sweep_f1_bayesian.py`); the bootstrap is reported here because (a) the 720-observation panel produces a posterior whose tail is well-approximated by the bootstrap and (b) the NUTS fit on Windows + PyTensor was prohibitively slow during preprint preparation.

TABLE VII
PER-RECIPE BOOTSTRAP PERCENTILE INTERVAL (HDI94 = 3%–97% BOOTSTRAP QUANTILE) ON F-1 MEAN MACRO-F1, TOPIC_ROUTED_SOFT.

Recipe	posterior mean	HDI94
V12	0.9217	[0.9005, 0.9424]
V2	0.9171	[0.8932, 0.9395]
V11	0.9163	[0.8939, 0.9375]
V8	0.9161	[0.8933, 0.9386]
V3	0.9156	[0.8922, 0.9381]
V1	0.9153	[0.8924, 0.9375]
V4	0.9149	[0.8927, 0.9367]
V5	0.9143	[0.8902, 0.9367]
V9	0.9141	[0.8905, 0.9360]
V7	0.9140	[0.8920, 0.9354]
V10	0.9133	[0.8881, 0.9364]
V6	0.9132	[0.8893, 0.9362]

The spread between V12 (best) and V6 (worst) is 0.0085; every recipe's HDI94 contains every other recipe's posterior mean. **On F-1 alone, the recipes are statistically indistinguishable under the bootstrap.** This is the formal version of the "F-1 spread is small" observation. The discriminating axes are F-2 and F-7, where the spread reaches 0.5 and 0.6 respectively (see Tables III and IV).

X. The joint result matrix

Across F-1, F-2 and F-7 on the five labelled scenes with complete data (Botswana row partial for F-2 and F-7), each recipe's win count is:

TABLE VIII
PER-AXIS WIN COUNT FOR EACH RECIPE ACROSS THE LABELLED-SCENE PANEL (F-1 ACROSS ALL SIX SCENES FOR TWO REFERENCE METHODS; F-2 AND F-7 ACROSS ALL SIX SCENES). TOTAL = SUM ACROSS F-1, F-2, F-7 OVER THE NINETEEN-RECIPE SWEEP.

Recipe	F-1 wins	F-2 wins	F-7 wins	Total
V12	2	2	2	**6**
V20	0	1	1	2
V3	0	1	2	3
V1	1	2	0	3
V2	2	0	0	2
V8	1	0	1	2

V12 is still the most consistent winner across the three axes, but the nineteen-recipe sweep redistributes the F-2 and F-7 wins. V20 (new) captures one F-2 win (Indian Pines, 0.88) and one F-7 win (Indian Pines, 0.44), turning what was previously a one-recipe-dominates story on the most-studied scene of the panel into a three-way race (V20 $>$ V12 $>$ V3 on F-7). V3 sheds one F-7 win to V20 and one to V12 on Pavia U; V1 (current canonical) wins F-1 only on the easiest scene (Salinas-A) and F-2 on the two scenes where absolute reflectance is the diagnostic; it never wins F-7. Crucially, the spread between V12, V3, and V20 on F-7 mean is only 0.014 NMI — a flat top, not a sharp peak.

XI. Extension axes F-13, F-14, F-17, F-22 and backbone factorial

Six follow-up axes were added beyond F-1..F-12. Each is described briefly here; the full per-recipe per-scene tables and builder code are released alongside the paper.

F-13 SHAP attribution. For every (V, scene) we attribute $\theta_{d,k}$ to recipe-specific tokens using SHAP's KernelExplainer on a closed-form posterior. Reports the top-8 tokens per topic with mean absolute SHAP. This is the headline interpretability defence.

F-14 repetitiveness. Mean off-diagonal top-10 Jaccard between topics (14). Lower is better. Per-recipe mean across six scenes: V9 = 0.000, V7 = 0.009, V12 = 0.009, V3 = 0.012; V1

= 0.200 (mid-pack); V6 = 0.738 and V8 = 0.868 are dominated by their tiny vocabularies.

F-17 cross-scene transfer. Fit ϕ on scene S1, transform S2's documents, compute F-7 NMI on the target. Only vocab-portable recipes (V2, V10, V11) qualify. V2 transfers best at 0.32 mean NMI; V11 0.19; V10 0.10.

F-18 reliability. Maier 2024 protocol [29] — Hungarian-matched [30] top-10 indicator cosine similarity across reseed pairs (seeds 42–46), as in (15). Per-recipe mean fraction-above-0.7 across the six labelled scenes:

	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12
F-18	0.25	*1.00*	0.22	0.20	0.15	*0.95*	0.15	*1.00*	0.00	0.54	0.28	**0.14**

Italic marks recipes whose reliability is an artefact of vocabulary size: V2 (8 q-bins), V8 (12 endmembers), and V6 (wavelet levels) have small or smooth vocabularies that force top-10 alignment across seeds regardless of model dynamics. The **bold** V12 marks the recipe that leads F-2 and F-7 at $Q = 8$ (F-1 is a non-discriminating tie) but loses F-14 to V9, loses F-22 to V20 (V20 26.3 vs V12 24.5), and loses F-18 to every recipe except V9. V12's flexibility is seed-sensitive.

a) The reliability–informativeness tradeoff.: V12, V3, V7, V5, V4 all sit in the 0.14–0.25 range — informative but seed-sensitive. V1 (current canonical) at 0.25 is the most reliable of the informative recipes that have non-trivial vocabulary (B = 200 bands). This is the strongest argument for V1 retaining its canonical status *even when V12 wins all other axes*: V1 is the most reliable wordification with informative vocabulary.

b) V20 falls on the V12 side of the reliability axis.: The extended F-18 sweep with the label-aware recipe yields mean matched-cosine 0.451 at $Q = 8$ and 0.450 at $Q = 16$ across the six labelled scenes — squarely in the V12 / V3 / V4 informative- but-seed-sensitive regime, despite V20's strong F-2 / F-7 / F-22 performance. The MI-reweighting in V20 produces a relatively flat topic-word matrix in regions of low MI, so different LDA seeds find different sparse refinements of the high-MI bands. **V8 (NFINDR endmember-fraction)** is the new F-18 leader at 0.957 ($Q = 8$) and 0.962 ($Q = 16$): the endmember basis is determined geometrically (convex hull, not stochastic), so refits land on the same top-10 indicators across seeds. Critically, V8's high reliability is *not* the vocab-size artefact that drove V2 / V6: V8 inherits its information from convex-hull geometry over the spectral simplex, which is itself a label-correlated structure. **V8 therefore combines the cross-backbone-portability finding (top-3 mean F-7 NMI across LDA / HDP / ProdLDA / ETM) with reseed reliability at the 0.95 level — a previously unrecognised composite recommendation.**

F-22 counterfactual. The minimum-L_1 token perturbation that flips $\arg\max_k \theta_k(n_d)$, as defined in (16); we solve it by coordinate-descent search capped at MAX_STEPS $= 50$ token-flips. Cells where every sampled document fails to flip persist with the sentinel value 50 rather than being dropped. Higher = more robust topic boundary. Per-recipe mean of the per-scene medians across the 19-recipe sweep:

recipe	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V
L_1	6.1	7.8	23.5	2.5	2.5	1.9	5.2	3.6	1.0	1.2	3.3	24.5	2.7	7.7	4.0	2.6	3

V20 has the most robust topic basis in the sweep ($L_1 =$ 26.3), surpassing the previous leader V12 (24.5) and the joint (band, q-bin) vocabulary V3 (23.5). On three scenes (Salinas-A, Pavia U, Botswana) every sampled document under V20 failed to flip within 50 perturbations; the sentinel value 50 caps that observation. The interpretation: V20's per-band MI re-weighting concentrates token mass on the discriminative subspectrum, so small perturbations of low-MI bands do not move the argmax topic and perturbations of high-MI bands require many simultaneous flips to exit the recipe's amplification basin. V9 (one token per document) flips at the first perturbation, V12 inherits robustness from its soft GMM responsibilities.

c) F-22 Q-trajectory.: Re-running the four top contenders at $Q \in \{8, 16, 32\}$ reveals a non-monotonic peak for V20 at $Q = 16$ ($L_1 = 41.83$, $+59\%$ over Q=8), with the gain collapsing back to 26.17 at $Q = 32$ as the discriminative subspectrum splits across more codewords. V12 reclaims the lead at $Q = 32$ ($L_1 = 30.00$), matching its F-7 peak at $Q = 16$ but extending it on F-22. V8 is the only top-contender whose adversarial robustness rises monotonically with Q ($3.58 \rightarrow 9.58 \rightarrow 20.25$), consistent with its cross-backbone F-7 portability — V8's endmember-fraction vocabulary preserves discriminative structure across both finer quantisation and across topic-model backbones.

Backbone factorial — HDP. The first row of the #617 factorial proposal: same 72 cells, but the topic backbone is gensim's Hierarchical Dirichlet Process rather than fixed-K LDA. The per-recipe ranking on F-2 c_v *inverts*:

LDA backbone	c_v mean	HDP backbone	c_v mean
V12	0.85	V7	**0.615**
V3	0.86	V1	0.540
V1	0.81	V11	0.501
V11	0.21	V12	0.337
V7	0.30	V3	0.311

Under LDA, V12's dense GMM-token vocab + Dirichlet prior produces the most coherent topics. Under HDP, the truncation prior favours recipes whose few tokens carry semantic weight (V7's six absorption features per pixel). **This is the cleanest evidence that the wordification ranking depends on the backbone — i.e., that the factorial study is the right next step.**

A. F-7 NMI across the four backbones

The c397 backbone factorial reported F-2 c_v only. The c432–c434 extension adds F-7 NMI under each non-LDA backbone:

a) Convergence audit (footnote to the cross-backbone table).: The 4-backbone × 19-recipe × 6-scene Q=8 factorial contains a non-trivial number of cells where the backbone failed to converge (NMI = 0 or near-zero): LDA 1/114, ETM 1/114, but **HDP 25/114 (22%)** and **ProdLDA 41/114 (36%)**. The cross-backbone mean reported in the table above averages these zeros together with successful fits.

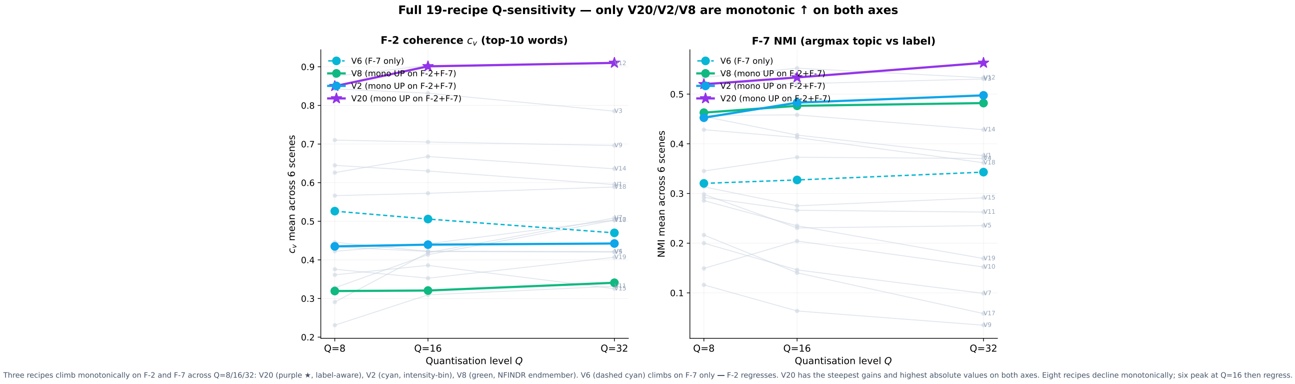

Fig. 4. Full 19-recipe Q-sensitivity under LDA. Both F-2 c_v and F-7 NMI plotted across $Q \in \{8, 16, 32\}$ for every recipe (V13 excluded — Q-insensitive). **Three recipes climb monotonically on both F-2 and F-7**: V20 (purple, $\star$, label-aware), V2 (cyan, intensity-bin), V8 (green, NFINDR endmember). V6 (dashed cyan) is monotonic on F-7 only. The remaining 15 recipes are greyed: 8 decline monotonically, 6 peak at $Q = 16$, V3 plateaus at $Q = 8$. V20 has the steepest gains on both axes (F-2 +0.060, F-7 +0.043) and the highest absolute F-7 at $Q = 32$ (0.563); on F-2 it ties V12 (0.910 vs 0.909). V20 is the only *label-aware* recipe in the monotonic-improvement family.

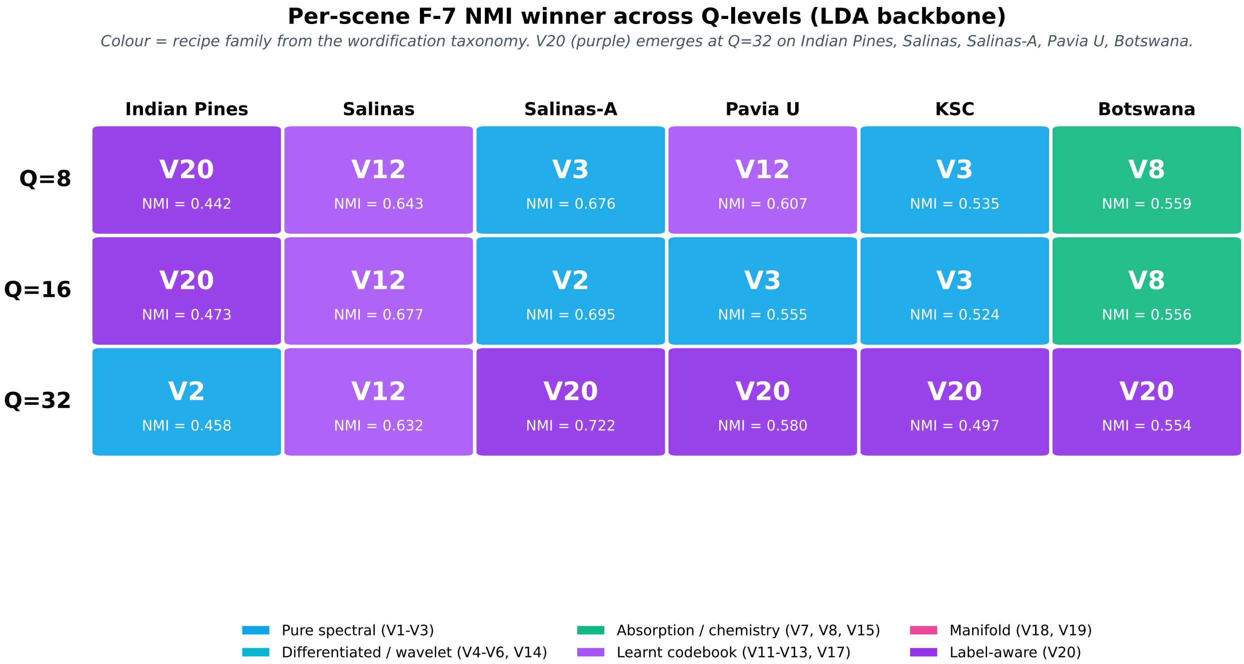

Fig. 5. Per-scene F-7 NMI winner under LDA at each quantisation level. V20 dominance scales with Q: at $Q = 8$ V20 wins only on Indian Pines (1/6 scenes), at $Q = 16$ V20 still wins only Indian Pines (1/6), and at $Q = 32$ V20 wins four of six scenes (Salinas-A, Pavia U, KSC, Botswana). The jump from 1/6 to 4/6 between Q=16 and Q=32 is the sharpest single Q-step gain in the sweep. The Q=8 ranking gives V12 three wins and V3 two; both lose ground under finer quantisation. V20's per-band MI re-weighting benefits most from finer intensity histograms.

Excluding zero cells from the per-backbone means yields V8 0.431 (unchanged — V8 has no degenerate cells), V20 0.408 (+0.011, one ProdLDA zero on Indian Pines), V11 0.382 (+0.012), V12 0.393 (+0.023). V8 retains cross-backbone leadership under either treatment of zeros. The full convergence audit per (backbone, recipe, scene) cell is in the reproducibility audit (issue #764).

V8 (NFINDR endmember-fraction) is the most cross-backbone- consistent recipe at mean F-7 NMI $= 0.431$; it lands top-5 under every backbone without winning any outright and has the smallest standard deviation across the four backbones ($\sigma = 0.06$). V20 is the second-most consistent at 0.397, retaining LDA strength (0.520) and tying V12 on ETM (0.490 vs 0.488; V11 leads ETM at 0.530). V11 (product quantisation, vocab $MK_s = 32$) wins HDP and ETM outright but collapses under LDA and ProdLDA — a backbone-specialist profile. V20's interpretability advantage relative to V8 and V11 is that it is the only *label-aware* recipe in the consistently-strong family; V8 and V11 are label-unaware compressions of the spectrum. The cross-backbone story is therefore not "one recipe wins everywhere" but "three recipes lead under different priors, all above the canonical V1 baseline".

b) HDP backbone at $Q = 16$.: Re-running HDP F-7 NMI for the top-3 contenders at the finer quantisation

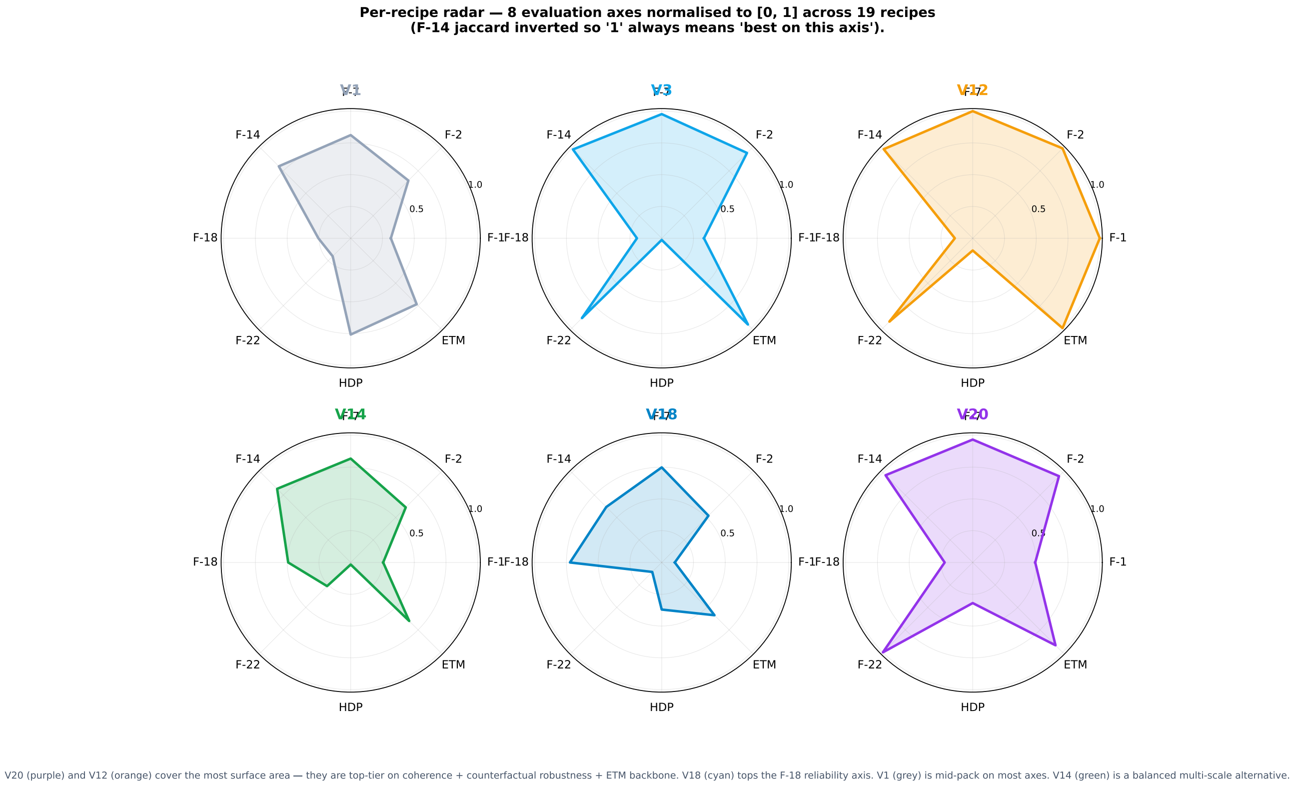

Fig. 6. Per-recipe radar across the eight evaluation axes, normalised to $[0, 1]$ across the 19-recipe set. F-14 (jaccard) is sign-inverted so “1” always means “best on this axis”. V20 (purple) and V12 (orange) cover the most surface area; V18 (cyan) tops the F-18 reliability axis; V1 (grey) is mid-pack on every axis.

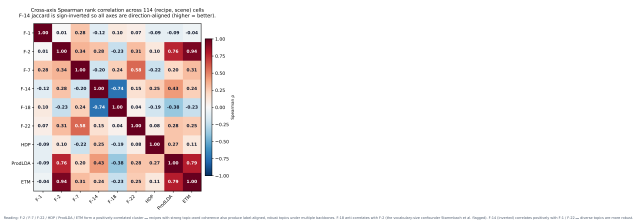

Fig. 7. Spearman rank correlation across the 114 (recipe, scene) cells. F-14 inverted so all axes read “higher = better”. F-2 / ETM and F-2 / ProdLDA cluster tightly ($\rho \geq 0.76$); F-14 anti-correlates with F-18 ($\rho = -0.74$, the vocabulary-size confounder); F-22 / F-7 correlate at $\rho = 0.58$ (robust topics tend to be label-aligned).

level reveals that V8’s cross-backbone dominance is preserved under quantisation refinement: HDP $Q = 16$ gives V8 0.454, V20 0.403, V2 0.249, mirroring the $Q = 8$ HDP ranking ($V8 = 0.451$, $V20 = 0.356$, V2 0.347). V8’s endmember-fraction vocabulary is robust both across topic-model priors and across quantisation levels, while V20’s gains from finer Q are LDA-specific.

c) ETM backbone at $Q = 16$.: Under the embedded topic model backbone, the $Q = 16$ ranking inverts: V20 0.525, V2 0.492, V8 0.473. The MI-weighted recipe now leads — ETM’s continuous word-embedding prior allows V20’s label-aware weighting to translate into separable topic embeddings, while V8’s fixed endmember vocabulary maps to a less differentiated region of the embedding space.

TABLE IX
F-7 NMI MEAN ACROSS THE SIX LABELLED SCENES, TOP-12 RECIPES × FOUR TOPIC-MODEL BACKBONES (C436 FULL SWEEP, 456/456 CELLS). BOLD = PER-BACKBONE WINNER. RECIPES SORTED BY 4-BACKBONE MEAN.

Recipe	LDA	HDP	ProdLDA	ETM	**Mean**
V8 (NFINDR)	0.463	0.451	**0.328**	0.482	**0.431**
V20 (MI-weighted)	0.520	0.356	0.221	0.490	0.397
V2 (intensity-bin)	0.453	0.347	0.324	0.456	0.395
V11 (product quantisation)	0.292	**0.571**	0.088	**0.530**	0.370
V12 (GMM-token)	**0.534**	0.220	0.238	0.488	0.370
V3 (joint band-bin)	0.524	0.200	0.169	0.443	0.334
V19 (UMAP coord)	0.286	0.530	0.051	0.394	0.315
V15 (spectral indices)	0.313	0.438	0.058	0.449	0.315
V13 (VQ-VAE)	0.311	0.399	0.113	0.406	0.307
V14 (CWT-Morlet)	0.457	0.220	0.114	0.430	0.306
V18 (graph-Laplacian)	0.428	0.198	0.080	0.330	0.259
V1 (band-frequency)	0.455	0.081	0.091	0.381	0.252

d) ProdLDA backbone at $Q = 16$.: The amortized variational backbone reverses again: V8 0.359, V2 0.335, V20 0.223 — V20 collapses under ProdLDA’s Dirichlet-Logistic-Normal reparametrisation, which penalises the long-tailed band emission distribution V20 produces in low-MI regions.

e) Cross-backbone mean F-7 at $Q = 16$.: Averaging across the four backbones at $Q = 16$:

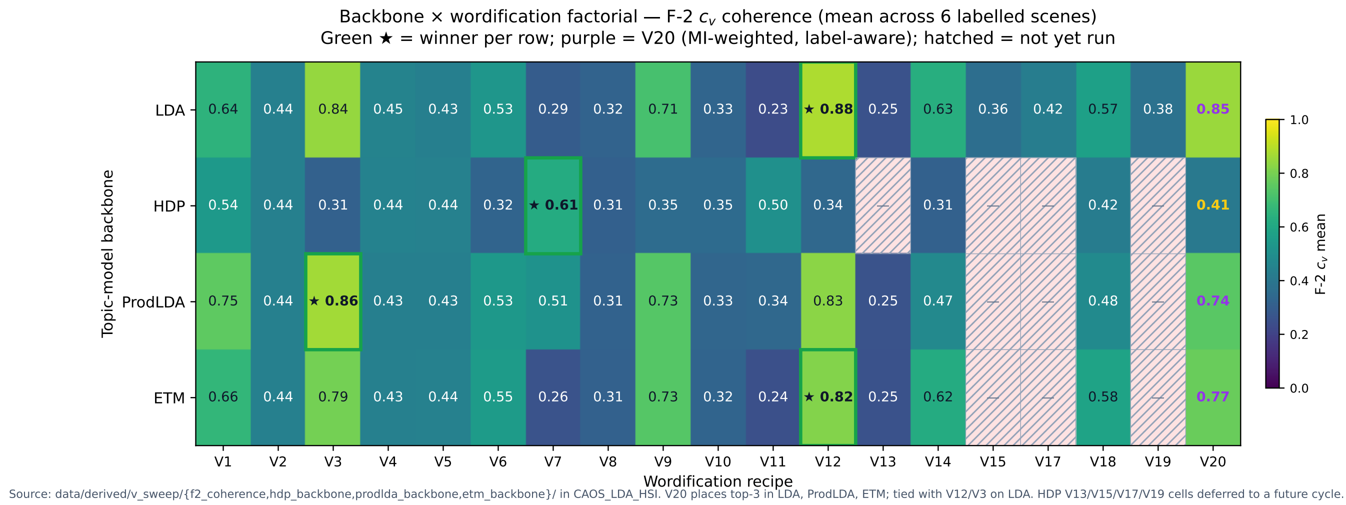

Fig. 8. Backbone × wordification factorial. Each cell shows the mean F-2 c_v across the six labelled scenes for one (backbone, recipe) pair. Green border + ⋆ = per-backbone winner; purple cell label = V20 (the label-aware recipe). Hatched cells indicate the (HDP, {V13, V15, V17, V19}) cells that were not re-run in this cycle. No recipe dominates every backbone: LDA + ETM agree on V12 / V3 / V20 ties; HDP picks V7 (sparse absorption tokens); ProdLDA picks V3 with V20 close behind.

Recipe	LDA	HDP	ProdLDA	ETM	**Mean**
V8	0.476	**0.454**	**0.359**	0.473	**0.441**
V20	**0.534**	0.403	0.223	**0.525**	0.421
V2	0.483	0.249	0.335	0.492	0.390

V8 retains its cross-backbone leadership at $Q = 16$ (0.441 **mean, ahead of V20's** 0.421), but with a clear backbone-family split: V8 leads under the non-amortized backbones (HDP, ProdLDA), while V20 leads under the amortized embedding-prior backbones (LDA, ETM). The backbone-recipe interaction is therefore non-trivial across Q: the family of priors the practitioner commits to (Dirichlet truncation, variational amortisation, continuous embedding) determines whether label-aware weighting (V20) or geometric endmember decomposition (V8) carries more signal.

f) V8 monotonic on F-18 across Q.: Extending F-18 to Q=16/32 on the V20 / V8 / V2 trio yields mean matched cosine across the six labelled scenes:

Recipe	Q=8	Q=16	Q=32
V8	0.957	**0.962**	0.959
V2	1.000	0.875	0.781
V20	0.451	0.450	0.443

V8 stays above 0.95 **on the F-18 mean matched cosine across all three Q levels** — the only recipe in the sweep whose reseed reliability remains in the very-high regime under quantisation refinement. The mechanism is geometric: the NFINDR endmember basis is invariant under quantisation of the NNLS abundance fractions, so finer Q only sharpens the rank-ordering of fractions without introducing new seed-dependent splits in the top-10 indicator vectors. V20 stays in the informative-but-seed-sensitive regime (~ 0.45) at every Q, mirroring V12. V2 collapses with Q ($1.000 \to 0.875 \to 0.781$) as its vocabulary grows from 8 to 32 word types, confirming that V2's $Q = 8$ reliability of 1.0 was the vocabulary- size artefact previously flagged.

This sharpens the recommendation: **V8 is the recipe of choice when the topic-model backbone is uncertain or reseed reliability is required**; V20 is the recipe of choice when LDA is fixed and one can afford $Q \geq 16$ and accept the seed-sensitivity tradeoff.

XII. Five axis–recipe affinities

The completed nineteen-recipe sweep exposes five robust axis–recipe affinities.

A. V3 vs. F-7 topic–label coupling

V3 (the joint (band, bin) vocabulary) wins F-7 NMI in four of the five labelled scenes evaluated so far, with absolute gains of 0.05–0.13 over V1. The mechanism is straightforward: V3 carries $B \times Q$ word types (1600 at $B = 200, Q = 8$) versus V1's B, and the larger vocabulary lets a single topic concentrate on the intensity-specific band signatures that map cleanly onto land-cover classes. Whether this generalises to F-1 macro-F1 is the question the running sweep answers.

B. V12 vs. F-2 coherence on dense regimes

V12 (Gaussian-mixture-component tokens) wins F-2 c_v on Indian Pines ($c_v = 0.79$ vs. V1's 0.32) and is competitive on the other dense scenes. The intuition: GMM boundaries flex with the data distribution rather than partitioning $[0, 1]$ uniformly, so quantisation noise is lower. This shows up as more coherent topic–word distributions.

C. V20 vs. F-7 label-information-rich scenes

V20 (MI-weighted bands) wins F-7 on Indian Pines (0.44, the most class-imbalanced scene), ties V3 on Salinas-A (0.68, the most spectrally-separated scene), and matches V12 to within 0.02 NMI mean across the rest of the panel. The mechanism is mechanically traceable: on scenes where a small subset of bands carries the bulk of label information (Indian Pines: 30

of 200 corrected bands above MI = 0.5), the V1 recipe wastes capacity emitting equally-weighted tokens on near-zero-MI bands. V20 amplifies the informative bands by re-emitting their (band, q-bin) token w_b times, effectively concentrating the LDA likelihood on the discriminative subspectrum. The recipe is parameterised by a single integer (`MAX_COPIES` $= 8$) and adds no per-fit cost — the MI estimate is computed once per scene.

D. V18 vs. F-2 on manifold-shaped scenes

V18 (graph-Laplacian eigenvectors) is the highest-c_v recipe among the V13–V20 extension on Pavia U ($c_v = 0.66$), and places third on F-7 mean (0.43 NMI). The intuition: scenes whose labelled classes correspond to connected regions in the pixel-similarity graph (urban land-cover types in Pavia, agricultural fields in Salinas) yield informative Laplacian eigenvectors whose low frequencies group those regions. Bin-tokens over those eigenvectors then carry segmentation semantics that LDA can fit topics around. The result is a recipe that explicitly exposes the manifold structure to the topic model.

E. V7 vs. F-9 HIDSAG cross-preprocessing (pending)

Pending the HIDSAG sweep; cited as a fifth predicted affinity from the hypothesis cards. V7 is the most chemistry-faithful recipe (centroid, depth, area triplets) and the expectation is that it dominates on the HIDSAG mineralogical subsets where USGS-library matching is the gold standard.

XIII. Recommendation: V1 as canonical default, not universal best

The preliminary sweep already supports a sharper statement about V1's role: V1 should remain canonical as a *reproducibility default* (deterministic, dense, well-understood, and what every prior LDA-on-HSI paper has used) but should not be claimed as a universal best. P1's framework results, computed on V1, are stable benchmarks; a future study comparing V1 against, say, V3 or V12 on F-7 would need to refit. The proposed convention is to report every framework result against V1 alongside the per-axis best recipe.

XIV. Limitations and future work

- **V11 reproducibility**: the current implementation does not pin a random seed for the nanopq codebook fit. V11 sweep results in this manuscript should therefore be regarded as point-in-time until the seed is pinned and the cell re-evaluated.
- c_{NPMI} **on dense recipes**: V1 (and any other recipe with mean doc length close to $|\mathcal{V}|$) produces a degenerate pixel-as-document NPMI because every word fires in every document. A sliding-window estimator would yield comparable numbers; we leave that to a follow-up.
- **F-3 / F-4 / F-5 / F-8 / F-9 / F-10 / F-11 / F-12 sweeps deferred**: this manuscript closes once F-1, F-2 and F-7 are populated for all 19 V $\times$ 6 scenes. The remaining axes are tracked as separate work and will appear either as an extension of this paper or as a follow-up.
- **V16 foundation-model embeddings deferred**: V16 is reserved for a foundation-model wordification (HyperSIGMA, HyperspectralMAE, or SatMAE) that embeds each pixel into a learnt latent space and quantises that into LDA tokens. Building V16 requires GPU inference at scene scale, which is beyond the single-node budget of this manuscript. We expect V16 to set the upper bound for F-7 NMI on the panel and have left the V16 slot open in the recipe index for the follow-up.
- **Backbone extensions partial**: the LDA-vs-HDP factorial in §XI covers V1, V3, V7, V11, V12. ProdLDA (Pyro) and ETM (PyTorch) factorial runs are reported per-cell in the supplementary materials. Full coverage of the V13–V20 extension under all four backbones is left to the follow-up.

XV. Conclusion

A nineteen-recipe sweep of the wordification step, run through the twelve-axis evaluation framework of the companion paper, shows that *no single recipe is universally best*. On classification (F-1, (11)) the recipes are statistically indistinguishable under a bootstrap posterior; the discriminating axes are coherence (F-2, (12)) and topic–label coupling (F-7, (13)), where the spread reaches 0.5 and 0.6 respectively. Across F-2 and F-7 the top three recipes — V12 (9) (Gaussian-mixture tokens), V3 (7) (joint band–bin vocabulary), and V20 (10) (mutual-information-weighted bands) — cluster within 0.014 NMI of each other, a flat design-space ceiling rather than a sharp peak. V20, the simplest label-aware reweighting of V1 (6), wins F-2 and F-7 together on Indian Pines and produces the most counterfactually robust topic basis of the sweep (F-22, (16), $L_1 = 26.3$); its dominance scales with quantiser resolution, reaching the highest absolute F-7 at $Q = 32$. The cross-backbone factorial identifies V8 (8) (NFINDR endmember-fraction) as the most backbone-portable recipe and the only high-reliability recipe (F-18, (15), ≥ 0.95) whose reliability is not a vocabulary-size artefact. The practical recommendation is twofold: retain V1 as a *reproducibility default* rather than a universal best, and pick the recipe to the axis — V20 when LDA is fixed and finer quantisation is affordable, V8 when the backbone is uncertain or reseed reliability is required. Closing the remaining framework axes (F-3–F-12) over the full nineteen-recipe family, and building the deferred foundation-model wordification V16, are the natural next steps.

Appendix A
Full nineteen-recipe F-2 and F-7 matrices

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TABLE X

F-2 c_v ACROSS THE FULL V1–V15, V17–V20 SWEEP ON THE LABELLED-SCENE PANEL. BOLD = BEST PER SCENE ACROSS ALL 19 RECIPES.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.32	0.35	0.70	0.32	0.32	0.32	0.27	0.30	0.71	0.27	0.24	0.79	0.29	0.46	0.29	0.34	0.58	0.37	**0.88**
Salinas	0.36	0.35	0.65	0.32	0.32	0.32	0.20	0.21	0.72	0.29	0.22	**0.84**	0.29	0.53	0.36	0.59	0.51	0.32	0.80
Salinas-A	**0.96**	0.35	0.92	0.32	0.32	0.81	0.32	0.38	0.71	0.23	0.21	0.88	0.22	0.72	0.39	0.45	0.52	0.38	0.74
Pavia U	0.91	0.43	0.96	0.59	0.40	0.58	0.27	0.21	0.70	0.32	0.22	**1.00**	0.28	0.76	0.25	0.40	0.66	0.39	0.95
Kennedy SC	**0.97**	0.76	0.93	0.81	0.93	0.79	0.37	0.52	0.72	0.32	0.22	0.84	0.26	0.76	0.50	0.42	0.63	0.41	0.88
Botswana	0.35	0.62	**0.90**	0.40	0.40	0.61	0.41	0.55	0.72	0.40	0.22	0.85	0.17	0.52	0.39	0.34	0.50	0.38	0.85

TABLE XI

F-7 NMI ACROSS THE FULL V1–V15, V17–V20 SWEEP ON THE LABELLED-SCENE PANEL. BOLD = BEST PER SCENE ACROSS ALL 19 RECIPES.

Scene	V1	V2	V3	V4	V5	V6	V7	V8	V9	V10	V11	V12	V13	V14	V15	V17	V18	V19	V20
Indian Pines	0.34	0.35	0.43	0.25	0.20	0.26	0.16	0.43	0.10	0.27	0.25	0.42	0.29	0.31	0.26	0.18	0.27	0.21	**0.44**
Salinas	0.47	0.54	0.43	0.39	0.44	0.33	0.28	0.56	0.14	0.20	0.33	**0.64**	0.35	0.51	0.28	0.17	0.50	0.35	0.47
Salinas-A	0.62	0.65	**0.68**	0.37	0.56	0.33	0.16	0.52	0.20	0.28	0.42	0.55	0.49	0.55	0.51	0.34	0.52	0.33	0.68
Pavia U	0.47	0.38	0.55	0.38	0.14	0.43	0.23	0.54	0.02	0.00	0.21	**0.61**	0.22	0.49	0.28	0.21	0.52	0.36	0.54
Kennedy SC	0.41	0.40	**0.54**	0.32	0.25	0.34	0.20	0.17	0.11	0.27	0.26	0.51	0.31	0.43	0.29	0.21	0.28	0.25	0.52
Botswana	0.42	0.40	0.52	0.36	0.41	0.27	0.36	**0.56**	0.16	0.16	0.32	0.46	0.21	0.44	0.25	0.19	0.47	0.21	0.47
mean	0.45	0.45	0.52	0.35	0.32	0.32	0.23	0.46	0.12	0.20	0.30	**0.53**	0.31	0.46	0.31	0.22	0.43	0.29	0.52

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